Ankrit Jarngal

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Education

National Institute of Technology Srinagar, Jammu & Kashmir, India Aug 2023 – May 2027
B.Tech. in Computer Science & Engineering CGPA: **8.58**

Experience

Full Stack Engineer Intern, IIT (BHU) Varanasi [Remote] Oct 2025 – Jan 2026

- Designed and developed **CodeJudge**, a Smart Code Editor integrated with an **LMS**, featuring **multi-language execution (C, C++, Python, JS, Java)** via Dockerized sandboxes, a Monaco Editor frontend, and **bring-your-own-key** LLM support for institutional deployments.
- Engineered backend modules for automated evaluation and **strict proctoring** (tab-switch detection, progressive warnings, auto-lock after 3 violations) with instant auto-grading and a Socratic AI tutor that provides hints rather than answers.

Fullstack Development Intern, AsynzX Computing (Bangalore, India) [Remote] May 2025 – Nov 2025

- Engineered **3** production-grade websites, converting Figma designs into responsive React components and integrating Node.js + Express REST APIs and Python FastAPI, improving deployment efficiency by **30%**.

Generative AI Intern, AgentsMadeEasy LLC (Delaware, USA) [Remote] Mar 2025 – Apr 2025

- Conducted end-to-end testing of a no-code AI platform, identifying **30+** reproducible bugs and delivering UI/UX recommendations that improved usability, accessibility, and AI workflow accuracy.

Achievements

- **1st Place — India's First Cursor AI Hackathon, NIT Srinagar (Mar 2026):** Won **First Prize** among **130+ teams** (**650+ registrations**, **80+ projects** submitted; $10,000+ in credits on the line) for *UncDoIt* — a browser extension delivering step-by-step, voice-guided web navigation in **11 Indian languages**, under the theme **"Build for the Next Billion."**
- **2nd Place — Coding Challenge, IUST Pulwama:** Secured **2nd** place in a competitive programming and DSA-focused contest among **15+** teams from multiple institutes.
- **Hackathon Runner-Up, NIT Srinagar:** Achieved **2nd** place among **20+** teams for *Mind Space*, recognized for rapid prototyping and AI-driven features.

Projects

GopherVectra: Vector Database Engine in Go GitHub

- Built a vector database in Go implementing an **LSM-Tree** (WAL + Memtable + multi-level SSTable compaction) for durability and an **HNSW** graph for $\mathbf{O}(\log \mathbf{N})$ ANN search across **768-dimensional** vectors; engineered a custom **Bloom filter** (FNV-64a) per SSTable to eliminate unnecessary disk I/O.
- Implemented HNSW neighbor pruning (diversity heuristic), monotonic node ID counter, and **two-heap search traversal** replacing naive sort-on-every-iteration; achieved **100% recall** validated against brute-force flat scan.
- Reduced latency ∼**2000ms** → ∼**247ms** and throughput **5** → **250+ QPS** (**50**× improvement) with **10** concurrent users on **1000** vectors of **768** dimensions.

Dungeon Explorer: Real-Time Multiplayer Roguelike Game Engine GitHub — Live

- Developed a **turn-based multiplayer roguelike** with a concurrent Go backend managing multiple **5-player** sessions using goroutines and channels.
- Designed **procedural dungeons** with Random Walk and corridor-carving, generating fully unique, fully connected maps featuring loot, healing fountains, and guarded exits.
- Programmed AI monsters (Goblins, Ogres, Bats, Guardians) with **multi-state behavior**: wander, chase, attack, and return; incorporated Fog of War and shared vision mechanics for team strategy.
- Engineered **WebSocket** command protocol for synchronized turns and authoritative game state, handling **100+** concurrent commands per second.

MCP (Model Context Protocol) Assistant GitHub — Live

- Architected a **modular AI assistant** framework using **MCP** for multi-step query resolution and dynamic context enrichment.
- Designed **multi-model structure** where specialized LLMs handle intent recognition, summarization, and task automation; integrated **Google APIs** for scheduling, emails, and event management, achieving **90%** accurate intent recognition in testing.

Technical Skills

Languages: C, C++, JavaScript, TypeScript, Python, Go
Dev Tools: ReactJS, Next.js, Node.js, Express, Tailwind CSS, Git, Docker, Vercel
Databases: MongoDB, PostgreSQL, Pinecone (Vector DB)
AI/ML: NLP, Reinforcement Learning (Q-Learning), Prompt Engineering, Vector Search, Automation